

Project Report: Bitcoin Time Series Analysis and Volatility Forecasting

Problem Statement

Cryptocurrency markets, particularly Bitcoin, are known for extreme price swings that make them both attractive to traders and difficult to analyze reliably. Understanding whether Bitcoin's price movements follow any predictable pattern, and how its volatility behaves over time, is valuable for risk management, trading strategy design, and general market understanding. This project applies classical time series statistics to Bitcoin's historical daily price data to rigorously test both of these questions rather than assume the answers.

Project Overview

We built a time series analysis pipeline that models Bitcoin's daily log returns using ARIMA, and separately models the volatility (risk) of those returns using GARCH. The project follows a structured statistical workflow: cleaning and preparing the data, exploring it visually, selecting predictive features using proper statistical tests rather than intuition, verifying the assumptions required for time series modeling (stationarity, seasonality), fitting both models, and validating their results against independent benchmarks. The analysis was carried out in R.

Dataset Summary

The dataset (bitcoin_2010-07-17_2024-06-28.csv) contains daily OHLCV (Open, High, Low, Close, Volume) records for Bitcoin from 2010-07-17 to 2024-06-28 (5,095 rows). Key fields used:

- Start (Date)
- Open, High, Low, Close (daily price levels, in USD)
- Volume (daily trading volume)

The Market Cap column was excluded as it is mathematically derived from Close price and circulating supply, and therefore redundant. The data was checked for missing dates, missing values, and OHLC logical consistency, and was found to be clean. The core target variable, log return, was derived as $\log(\text{Close}_t / \text{Close}_{(t-1)})$, the standard transformation used in financial time series analysis.

Exploratory Data Analysis (EDA Highlights)

- Bitcoin's price could only be meaningfully visualized on a log scale, due to its six-order-of-magnitude range across the sample (\$0.05 to over \$70,000).
- Daily log returns showed clear fat tails compared to a normal distribution, confirmed by both a histogram overlay and a QQ-plot.
- A 30-day rolling volatility plot revealed strong volatility clustering: turbulent periods (2011, 2013-14, 2018, 2020) were followed by more turbulent days, and calm periods stayed calm.
- Raw Close price showed a clear non-stationary trend, while log returns appeared to fluctuate around a stable mean.

Model Development

Feature evaluation was performed statistically rather than by intuition: lagged returns (1, 2, 3, 5, and 7 days) were tested using correlation analysis and multiple linear regression.

Two models were then developed:

- ARIMA(2,0,3) - fit on log returns to model return direction, with differencing fixed at zero since returns were already confirmed stationary via the Augmented Dickey-Fuller (ADF) test.
- GARCH(1,1) - fit on the same log returns with a Student-t error distribution (justified by the fat tails observed in EDA) to model volatility clustering.

A one-way ANOVA test on day-of-week effects found no statistically significant seasonality ($p = 0.709$), confirming that a seasonal model (SARIMA) was not required, consistent with Bitcoin's continuous 24/7 trading structure.

Evaluation Metric

To evaluate model performance and robustness, we used:

- R-squared (R^2), to measure explanatory power
- RMSE and MAE, to measure out-of-sample forecast accuracy against a naive random-walk baseline
- Correlation against an independent, model-free realized volatility measure, to validate GARCH

ARIMA achieved a low R^2 (approximately 0.004 in-sample, -0.011 out-of-sample), reflecting Bitcoin's return direction being close to unpredictable - consistent with market efficiency. Despite this, ARIMA outperformed a naive baseline on RMSE (0.0297 vs 0.0421) and MAE (0.0204 vs 0.0305), since the naive method propagates noise by copying the previous day's return forward. In contrast, GARCH's alpha and beta coefficients were both highly significant ($p < 0.001$, alpha+beta approximately 0.999), a rolling volatility-persistence regression achieved $R^2 = 0.986$, and GARCH's estimated volatility correlated at 0.478 with an independent realized volatility measure - confirming volatility, unlike direction, is strongly and reliably predictable.

Implementation

The analysis was implemented as a structured R script, organized into clearly separated stages (data loading, EDA, feature selection, stationarity testing, ARIMA, GARCH, validation). Each stage produces its output - summary statistics, test results, and diagrams - saved directly to a dedicated project folder, so that every plot and result remains available and organized throughout the analysis rather than being lost between steps.

- Diagrams generated: Close price (log and linear scale), daily log returns, return distribution vs. normal curve, 30-day rolling volatility, QQ-plot, ACF/PACF of returns, and GARCH vs. realized volatility comparison.
- All diagrams were saved as PNG files to a single, dedicated diagrams folder for easy reference and inclusion in reporting.

Challenges

- Early Bitcoin trading data (pre-2013 for Volume, pre-2018 for OHLC range) contained artifacts from thin trading activity, requiring careful flagging rather than blind use, particularly when validating GARCH against realized volatility.
- An initial automated ARIMA order search over-differenced the already-stationary return series, producing an unstable model; this was identified through coefficient inspection and corrected by fixing the differencing order.
- Balancing statistical rigor with interpretability - ensuring each test's result (e.g., a low R^2) was explained in context rather than presented as a standalone number.

Impact

This analysis can:

- Serve as evidence that Bitcoin's short-term price direction is not reliably forecastable using its own historical returns, supporting a risk-management rather than a directional-trading use case
- Provide a validated volatility model (GARCH) usable for risk estimation, position sizing, or options-pricing-adjacent applications
- Serve as a learning template for applying and correctly interpreting classical time series statistical methods on financial data

Tech Stack

- R
- tidyverse / dplyr, ggplot2
- tseries (ADF test)
- forecast (ARIMA)
- rugarch (GARCH)
- zoo (rolling statistics)

Future Work

- Incorporate exogenous regressors such as trading volume (ARIMAX / GARCH-X) to test whether volume adds predictive power
- Explore asymmetric volatility models (EGARCH, GJR-GARCH) to test for leverage effects
- Implement full walk-forward backtesting for a more rigorous out-of-sample evaluation
- Explore deep learning approaches (e.g., LSTM) as a non-linear alternative for comparison